

Analytics Agent

Role: Data Aggregator · Pipeline Position: Step 4 of 5 · Endpoint: POST /analytics/report

Aggregates fleet metrics, computes efficiency and utilization, and generates recommendations.

Overview

The Analytics Agent is the fourth agent in the pipeline. It computes trip totals, fuel efficiency, utilization percentage, and maintenance frequency for a given vehicle, then applies rule-based logic to output an actionable recommendation.

Metric Formulas

Metric	Formula
Fuel Efficiency	$\text{total_distance} / \text{total_fuel}$ (km/L)
Utilization %	$\min(100, (\text{total_distance} / 17000) \times 100)$
100% threshold	17,000 km total distance = 100% fleet utilization

Recommendation Rules

Condition	Recommendation Output
utilization < 40	"Vehicle underutilized."
maintenance_count > 5	"Frequent maintenance detected."
fuel_efficiency < fleet_avg AND fuel > 0	"Vehicle fuel efficiency is below fleet average."
All other cases	"Vehicle operating normally."

Fleet average defaults to 8.0 km/L if no historical fuel data exists in the database.

API Endpoints

Method	Path	Description
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POST	/analytics/report	Generate a performance report for a specific vehicle
GET	/analytics/historical	Retrieve monthly trip-count trend data for charting

Historical Chart Logic

- Queries trips table grouped by month.
- Uses `strftime('%m', created_at)` on SQLite; `to_char(created_at, 'MM')` on PostgreSQL.
- Scales trip counts: `points[month] = min(100, max(50, 60 + trip_count × 8))`.
- Falls back to static baseline `[88, 72, 91, ...]` when no trip data exists.

Internal Logic Flow

```
generate_report(vehicle_id)
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    ■■■ 1. get_vehicle(vehicle_id)           → VehicleUnavailableException
    ■■■ 2. get_trip_statistics(vehicle_id)    → total_trips, avg_distance, total_fuel
    ■■■ 3. get_maintenance_statistics(vehicle_id) → maintenance_count
    ■■■ 4. get_fleet_average_fuel_efficiency() → fleet_avg km/L
    ■■■ 5. Compute fuel_efficiency and utilization
    ■■■ 6. Evaluate recommendation rules
    ■■■ 7. Return structured analytics dict
```

Module Files

File	Purpose
agent.py	Agent registry entry — wraps service call
service.py	Business logic: metric computation, utilization, recommendation rules
repository.py	DB queries: trip stats aggregation, maintenance count, fleet avg
routes.py	FastAPI router — POST /analytics/report + GET /analytics/historical
schemas.py	Pydantic request/response models
prompts.py	AI prompt templates
tools.py	LLM tool definitions